STANDARD OPERATING PROCEDURE (SOP) - SAMPLE FORMAT

BLOOD GLUCOSE ESTIMATION BY GOD-POD METHOD

SOP Code	________________________	**Version**	01
Department	Clinical Biochemistry	**Effective Date**	____ / ____ / ______
Prepared By	________________________	**Reviewed By**	________________________
Approved By	________________________	**Review Date**	____ / ____ / ______

Educational sample only. This document is provided as a sample SOP format. Before routine clinical use, the laboratory should adapt reagent-specific settings, incubation time, standard concentration, reference ranges, critical values, QC policy, analyzer settings, and other operational details to the approved manufacturer instructions and local laboratory policy.

1. PURPOSE:

To describe the standard procedure for the quantitative determination of glucose levels.

2. SCOPE & APPLICATION:

This SOP applies to all laboratory personnel performing glucose estimation in the clinical biochemistry laboratory.

3. RESPONSIBILITIES:

Registered laboratory personnel on duty.

4. PATIENT PREPARATION:

4.1 Fasting Blood Sugar (FBS)
Patient should fast for 8-12 hours; only water is allowed.

4.2 Post-Prandial Blood Sugar (PPBS)
Sample collected 2 hours after a standard meal.

4.3 Glucose Challenge Test (GCT)
No fasting required. Sample collected 1 hour after 50 g oral glucose intake.

4.4 Oral Glucose Tolerance Test (OGTT)
Patient should fast for 8-12 hours. Fasting sample collected before 75 g oral glucose administration, followed by timed sample collection.

4.5 Random Blood Sugar (RBS)
Sample may be collected at any time without fasting.

5. TYPE OF SAMPLE:

Serum / plasma samples.

5.1 Reject sample: Hemolyzed sample and lipemic sample.

6. TYPE OF CONTAINER OR ADDITIVE:

Grey Top Tube: Contains Sodium Fluoride (NaF) as an antiglycolytic agent and Potassium Oxalate as an anticoagulant.
Red/Yellow Top Tube: For serum (requires immediate centrifugation).

7. REQUIRED MATERIALS:

7.1 Equipment:
- Semi-automated analyzer
- Micropipette
- Test tubes

- Timer
- Water bath (37 C)

7.2 Reagent:
- Ready-to-use glucose reagent
- Standard solution
- Controls
- Distilled water

7.3 Storage: As per manufacturer instructions (usually 2-8 C).

8. SAFETY PRECAUTIONS:

- Follow universal precautions.
- Wear PPE.
- Handle all samples as infectious.
- Dispose of waste as per biomedical guidelines.

9. PRINCIPLE:

Glucose oxidase (GOD) oxidizes the specific substrate beta-D-glucose to gluconic acid and hydrogen peroxide (H_2O_2) is liberated. Peroxidase (POD) enzyme acts on hydrogen peroxide to liberate nascent oxygen (O_2), then nascent oxygen couples with 4-amino antipyrine and phenol to form red quinoneimine dye.

The intensity of the colour is directly proportional to the concentration of glucose present in the sample. The intensity of colour is measured by colorimeter at **510 nm** and compared with that of a standard treated similarly. Final colour is stable for at least 2 hours if not exposed to direct sunlight.

10. PROCEDURE: (SET UP IN SEMI-AUTOMATED ANALYZER)

Mode of reaction: **End-Point**
Slope of reaction: **Increasing**
Wavelength: **510 nm (490-550 nm)**
Incubation time: ________________ *(as per reagent manufacturer instructions)*
Zero setting with: **Reagent blank**
Sample volume: **0.01 mL (10 uL)**
Reagent volume: **1.0 mL**
Standard concentration: ________________ **mg%** *(as per reagent manufacturer instructions)*
Linearity: **600 mg/dL**

10.3 MANUAL ASSAY PROCEDURE:

Bring reagent and sample to room temperature.
Set the analyzer "Zero" with distilled water.
Incubate at 37 C for 5 minutes and measure absorbance at 510 nm.

11. CALIBRATION / STANDARDIZATION PROCEDURE:

A calibration/standardization should be performed when a new reagent is used, after reagent lot change, or when QC values are out of control limits.

12. QUALITY CONTROL:

Run **normal or pathological** liquid-stable control at the start of each day/shift, after reagent lot change, and following major instrument maintenance or replacement.

13. CALCULATION:

Glucose in mg% = (Absorbance of sample / Absorbance of standard) x Concentration of standard

14. NORMAL / REFERENCE RANGE:

- Fasting (FBS): 70-110 mg/dL
- Post-prandial (PPBS): 70-140 mg/dL

- GCT: <=140 mg/dL
- Random (RBS): <=140 mg/dL
- OGTT (75 g):
  1 hour: <=180 mg/dL
  2 hours: <=140 mg/dL

Manual Assay Setup

	Reagent Blank	Standard	Test
R1	1 mL	1 mL	1 mL
Standard	-	10 uL	-
Sample	10 uL (D/W)	-	10 uL (PS)

15. LIMIT OF DETECTION:

This test is linear up to 600 mg/dL.
Samples above 600 mg/dL should be diluted 1:2 with normal saline. In such case, the result obtained should be multiplied by the dilution factor to obtain the correct glucose value.

16. LIMITATION / INTERFERENCE:

- **Icterus:** High bilirubin may cause falsely low results.
- **Lipemia:** Grossly lipemic samples should be clarified or the patient re-sampled after fasting.
- **Ascorbic acid:** High vitamin C levels may interfere with the POD reaction, causing falsely low results.

17. INTERPRETATION:

- **Hyperglycemia:** Seen in diabetes mellitus, Cushing's syndrome, and acute stress.
- **Hypoglycemia:** Seen in insulinoma, starvation, or insulin overdose.

18. CRITICAL VALUES:

< 40 mg/dL or > 450 mg/dL

19. REFERENCE:

Tietz Fundamentals of Clinical Chemistry and Molecular Diagnostics.
WHO Guidelines on Drawing Blood: Best Practices in Phlebotomy.

Note: Incubation time and standard concentration are intentionally left as reagent-specific fields because these may vary by manufacturer.

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